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Description - This procedure applies to all Signa Horizon systems. Sections 4 & 5 require **two** people.

1- INTRODUCTION

The gradient filter is a field-replaceable unit (FRU), so if it fails, the entire unit is replaced (except for the clear lug covers).

2- TOOLS REQUIRED

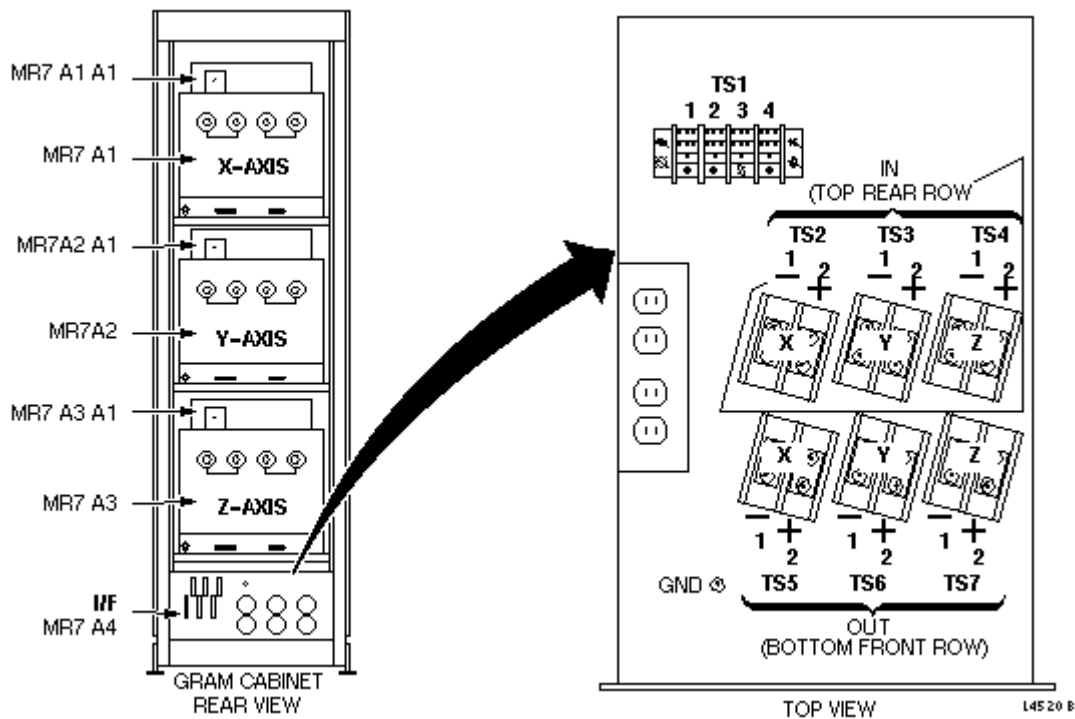
- 200–1200 in/lb torque wrench with 3/8-inch driver, 46-315668P2.
- Digital volt meter (DVM) with alligator clip leads

3- PRELIMINARY SET-UP

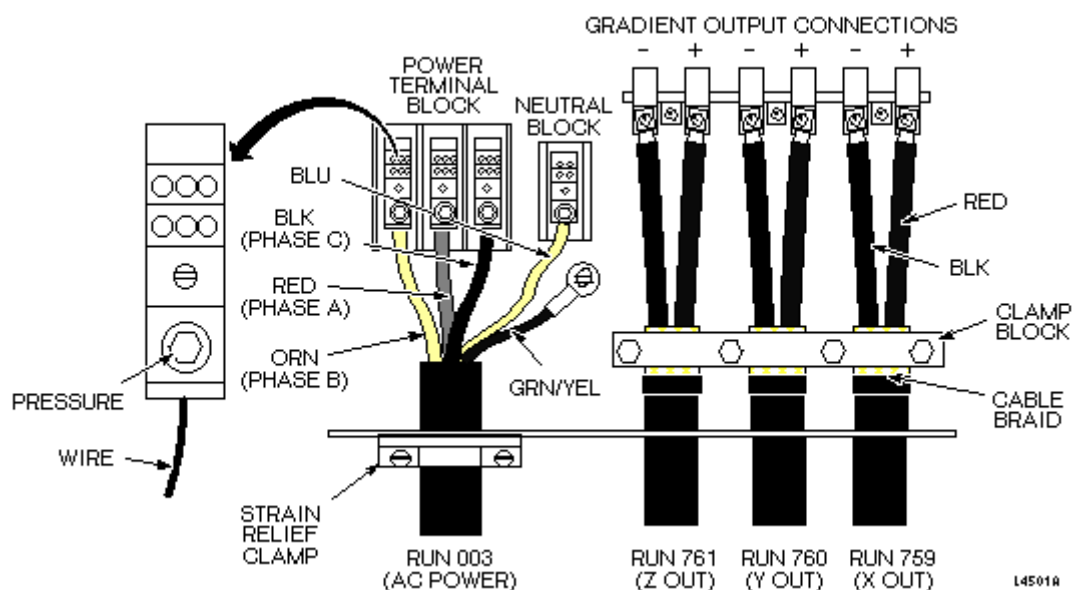


FATAL ELECTRIC SHOCK HAZARD!! THE GRAM AND GRADIENT AMPLIFIERS ACT AS CONSTANT LOAD SOURCES, AND WILL SEND MAXIMUM CURRENT TO ANY LOAD (INCLUDING YOU!). TO PREVENT FATAL ELECTRIC SHOCK, ENSURE THAT POWER IS OFF TO BOTH CABINETS BEFORE CONTINUING WITH THIS PROCEDURE.

1. Perform lock-out/tag-out by securing the PDU circuit breaker for the GRAM Cabinet and the 8645 Gradient Amplifier Cabinet with the lock-out/tag-out devices. Verify that all energy has been disabled by measuring incoming power to the GRAM cabinet at TS1 (see Illustration L4520B). Verify that all of the energy has been disabled for the 8645 Gradient Amplifier Cabinet by measuring power at TS1. Also see Illustration L4501A for the Signa Horizon HiSpeed system, or Illustration L4510A for Signa Horizon or Horizon EchoSpeed systems.

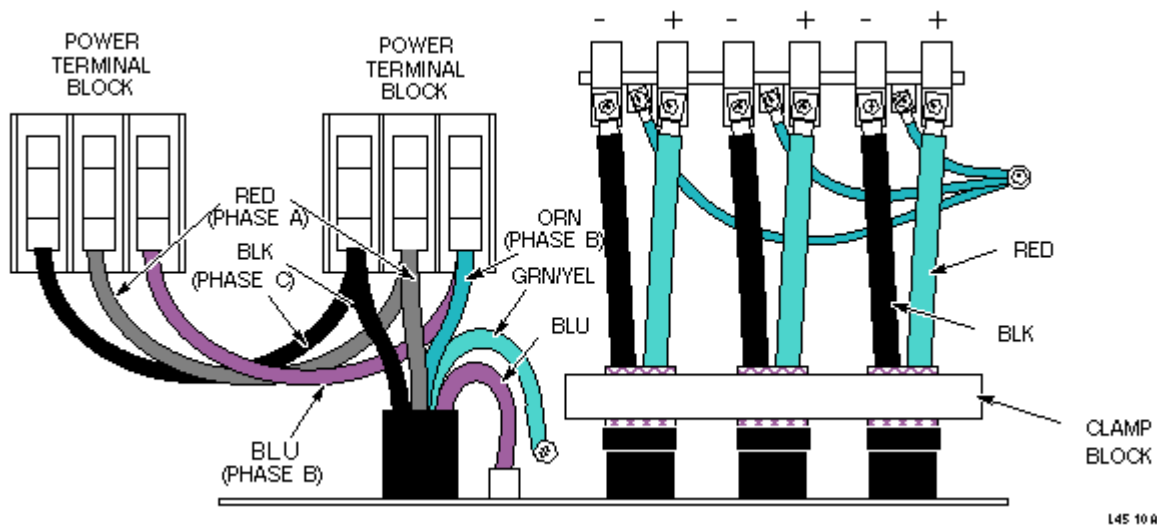


GRAM CABINET REAR VIEW - BOTTOM PANEL AND TS1
ILLUSTRATION L4520B



8645 CABINET POWER AND OUTPUT CABLE CONNECTIONS

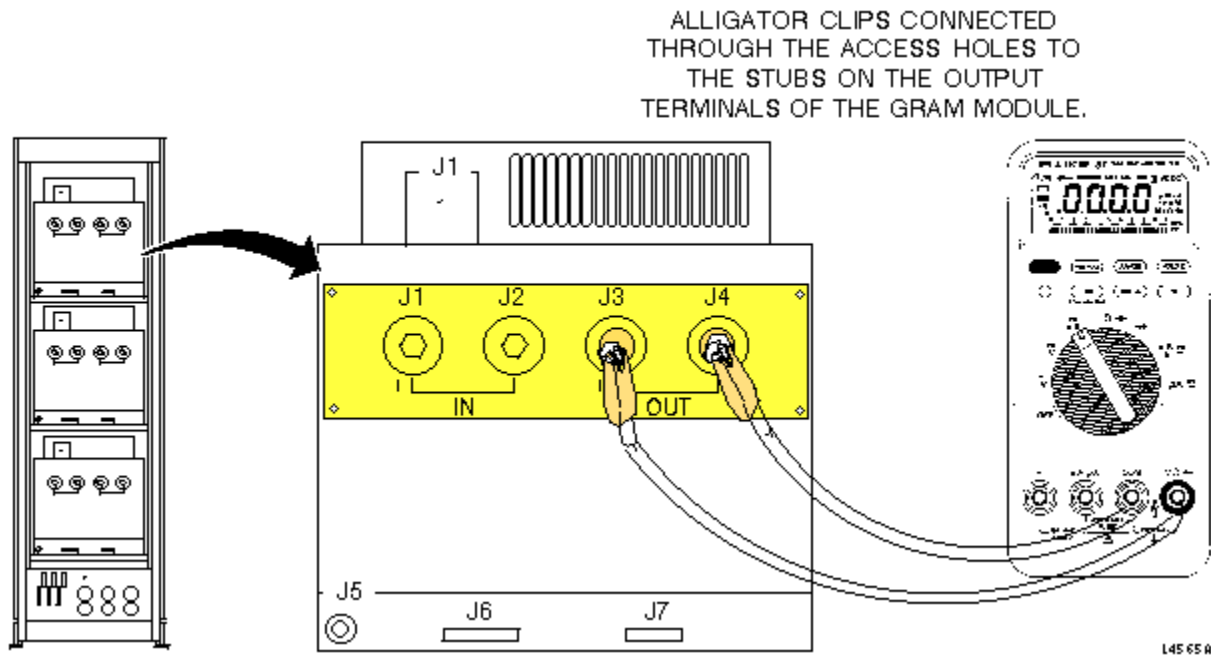
ILLUSTRATION L4501A



INCOMING POWER TO A DOUBLE-BAY 8645 GRADIENT CABINET

ILLUSTRATION L4510A

2. Verify that power is off by connecting a DVM across the input leads (J1, J2) and output leads (J3, J4) at rear of GRAM, as shown in Illustration L4565A.



GRAM: REAR VIEW SHOWING VOLTMETER ACROSS THE OUTPUTS
ILLUSTRATION L4565A

Note

When checking the input leads, you will have to hold the alligator clips against J1 and J2.

4- REMOVAL OF THE GRADIENT FILTER

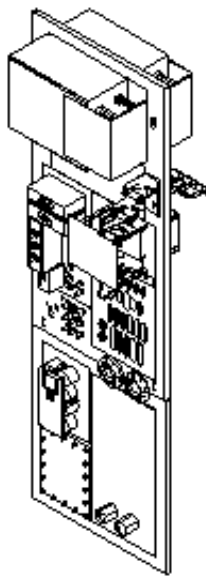
WARNING!

POSSIBLE PERSONAL INJURY AND/OR EQUIPMENT DAMAGE! TWO PEOPLE ARE REQUIRED FOR THIS STEP. ONE PERSON MUST BE IN THE EXAM ROOM TO HOLD THE FILTER AND KEEP IT FROM FALLING INTO THE EXAM ROOM, WHILE THE OTHER PERSON WORKS ON THE EQUIPMENT ROOM SIDE.



Two wrenches are necessary when removing the cables from the six GRAM input/output connectors on each side of the penetration panel: one wrench prevents the post from turning, while the other loosens the nut. If you try to loosen the nut with only one wrench, the post itself could turn, and damage to the circuit board inside the gradient filter could occur. See Illustration L4052A.

1. Using proper ESD precautions, follow the instructions in Illustration L4050A and Illustration L4053A.



L4050A

5- INSTALLING THE REPLACEMENT GRADIENT FILTER

1. With two people, install the replacement gradient filter per the instructions in Illustration L4054A, and Illustration L4055A.
2. Release the lock-out/tag-out from the PDU for the GRAM cabinet and the gradient cabinet.

6- SYSTEM RESTORATION

1. Perform Procedure for Geometry Verification next.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	April 17, 1998	W. Pasciak	Initial release converted to Word